

EL YOUSFI HAJAR

NETWORKS & TELECOMS ENGINEERING STUDENT (M2)

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Availability : 6-month Final-Year Internship (PFE) starting February 2027 — Casablanca / Rabat

PROFESSIONAL SUMMARY

Master's student in ISERT engineering (Networks & Telecoms, FSTM) with strong expertise across the telecommunications ecosystem from 2G to 5G (CDMA, O-RAN, xApp) and AI applied to networks. Rigorous and passionate about innovation, seeking a 6-month Final-Year Internship (PFE) in networks / telecommunications.

EDUCATION

FSTM – Master's in Embedded Systems, Networks and Telecommunications Engineering (ISERT) 2025 – 2027

- 2nd year (M2): networks & telecoms, embedded systems, artificial intelligence.

FSTM – Bachelor's in Computer Science and Multimedia Networks (IRM) 2024 – 2025

FSTM – DEUST in Computer Science 2022 – 2024

Ain Harrouda High School – Baccalaureate in Physical Sciences (French Option) 2021 – 2022

INTERNSHIP

Proactive Load Balancing preserving QoE using Deep Learning (O-RAN 5G) | FSTM Jun – Aug 2026

- Predictive load-balancing xApp for the O-RAN Near-RT RIC: LSTM/GRU prediction (92% accuracy) + DRL agent (PPO), with handover decisions through the E2 interface (E2SM).

Design of a Smart Intrusion Detection Camera (Raspberry Pi, Embedded Deep Learning) | FSTM Jun – Aug 2026

- Real-time people detection on Raspberry Pi 4 (PFA): YOLOv8n, 3-level alerts, GPIO (LED/buzzer) and MQTT publishing.

MENPS – IT Directorate (Bachelor Internship) | Rabat Apr – Jun 2025

- Development of the Mostaqbali application (React.js, Spring Boot) for the Ministry of National Education: application management, tracking, validation, securing data flows and automating the administrative process.

PROJECTS

Edge AI Telecom Gateway – Predictive maintenance of telecom sites | Python, FastAPI, WebSocket 2026

- Predictive maintenance of cellular towers: real-time analysis of telemetry (temperature, voltage, vibration) at the edge.
- Anomaly detection (Isolation Forest) + FastAPI REST API and WebSocket streaming; deployable on edge machines.

Cellular Network Planning and Optimization | Atoll 2026

- Radio planning of a cellular network with Atoll: propagation modeling, dimensioning and coverage analysis.
- Capacity and interference optimization; multi-band 4G/5G scenarios.

IP Network Automation and Monitoring Platform | Python, Grafana 2026

- Automation and monitoring platform for an IP network: metrics collection and automated configuration of network equipment.
- Grafana dashboards and real-time alerting for availability and performance monitoring.

TECHNICAL SKILLS

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| • Telecoms & RF : 2G, 3G, 4G, 5G, LTE, CDMA (BER/SNR), FTTH, GPON, Li-Fi, RF mixer 2.4 GHz CMOS | • AI & Data : Python, PyTorch, scikit-learn, OpenCV, LSTM/GRU, PPO, Anomaly Detection, Predictive Maintenance |
| • Mobile Networks & O-RAN : Near-RT RIC, xApp, E2, O-RAN, handover, QoE, Traffic Prediction | • Tools & Simulation : MATLAB, Simulink, ADS, CST, Atoll, Packet Tracer, Docker, Grafana, Git, LaTeX |
| • Data Networks : TCP/IP, VLAN/Trunk, STP, QoS, ACL, OSPF, Linux, GNS3 | • Security : Wireshark, Firewall, IDS, VPN |

CERTIFICATIONS

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| • CCNA Network Security – Cisco | • Claude AI – Anthropic |
| • CCNA Introduction to Network Security – Cisco | • Microsoft AI Classroom – Microsoft |
| • Network Defense – Cisco | |

LANGUAGES & ACTIVITIES

- **French** : Fluent, **English** : Fluent, **Arabic** : Native, **Italian** : Beginner, **Spanish** : Beginner, **Chinese** : Beginner
- **Activities** : Open Data Club (open data and analysis projects, FSTM 2026); Robothings Club (robotics and embedded electronics, workshops and tech competitions, FSTM 2023 – Present).